

Answer Key: Mid Semester 2

Instructions for TAs: This key provides the expected answers and the mark allocation for each question. Use your judgment for partially correct answers, awarding partial marks where appropriate.

Question 1 (Total 7 marks)

a. Which of the two trees (a) or (b) show the correct order of evolution from primitive to advanced forms? (2 marks)

- **Answer:** Both trees (a) and (b) are identical in terms of evolutionary relationships. They represent the same phylogram with nodes rotated, which does not change the branching pattern or the inferred relationships.
- **Reasoning:** The very concept of an “order from primitive to advanced forms” is biologically meaningless. All extant species have evolved for the same amount of time since the common ancestor and are therefore equally “advanced.” Evolution is not a linear ladder of progress; it is a branching process. Since both trees depict the same branching pattern, neither is more correct than the other in representing evolutionary relationships.
- **Marking:**
 - 1 mark for stating that both trees show the same relationships (or that neither is “correct” in a linear progressive sense).
 - 1 mark for explaining that the concept of primitive-to-advanced is not valid in evolutionary biology, or that node rotation does not change the tree.

b. Identify three clades in the tree (a) (3 marks)

- **Answer:** A clade is a monophyletic group consisting of an ancestor and all of its descendants. The following are examples of acceptable answers (naming conventions may vary). Any other valid clade (an ancestor and all its descendants) is also acceptable:
 1. The clade containing **Human and Cat** and their common ancestor T.
 2. The clade containing **Lizard and Duck** and their common ancestor S.
 3. The clade containing **Human, Cat, Lizard, Duck** and all their common ancestors, including R.
 4. Any clade containing a common ancestor and **ALL** its descendants.

- **Marking:** 1 mark for each correctly identified clade. A clade must be a monophyletic group (an ancestor and all its descendants).

c. Tree (b) shows that Frogs first evolved from sharks and then the lizards evolved from frogs. Comment on this statement. (2 marks)

- **Answer:** This statement is incorrect. The phylogenetic tree does not imply direct descent (i.e., that lizards evolved *from* frogs). Instead, it shows that frogs and lizards share a more recent common ancestor with each other than either does with sharks.
- **Marking:**
 - 1 mark for stating the statement is incorrect/false.
 - 1 mark for explaining the correct interpretation (common ancestry, branching process, not linear descent).

Question 2 (Total 3 marks)

Define (a) Red Queen Process (b) Adaptive Evolution (c) Disruptive Selection

- **(a) Red Queen Process (1 mark):** An evolutionary hypothesis where species must continuously adapt and evolve to survive in a constantly changing environment, often in the context of co-evolving species (e.g., predator-prey, host-parasite). The name comes from the phrase, “It takes all the running you can do, to keep in the same place.”
- **(b) Adaptive Evolution (1 mark):** The process by which a population becomes better suited to its environment over time through the increase in frequency of beneficial alleles due to natural selection.
- **(c) Disruptive Selection (1 mark):** A type of natural selection that favours extreme phenotypes over intermediate ones, leading to an increase in variance and potentially speciation.

Question 3 (Total 6 marks)

Using the Mexican tetra example, comment on:

a. Fitness is context specific (2 marks)

- **Answer:** In the surface habitat, having large eyes is adaptive for finding food and avoiding predators, making that phenotype fit. In the dark cave habitat, eyes are not only unnecessary but also energetically costly and prone to injury, so the eyeless phenotype has higher fitness. The same trait (eyes) is beneficial in one context and detrimental in another.

- **Marking:** 1 mark for explaining the surface context, 1 mark for explaining the cave context.

b. Evolution is not “progressive” (2 marks)

- **Answer:** The cavefish did not “progress” from an eyed ancestor to an eyeless form. Eyelessness is not a “more advanced” trait; it is a derived trait that evolved due to a specific environmental pressure. Evolution does not have a direction towards complexity. The loss of eyes is a simplification, showing that evolution can lead to trait loss when it is advantageous.
- **Marking:** 2 marks for a clear explanation that evolution is not directional and that loss/gain of a trait is not “progressive.” (1 mark for a partial explanation).

c. Trade-offs are common (2 marks)

- **Answer:** The loss of eyes in cavefish is associated with a trade-off. The energy and resources not used for developing and maintaining eyes are redirected to enhance other sensory systems (like smell, vibration detection, and brain areas for other senses). This improves fitness in the dark cave environment but would be a disadvantage in the surface environment.
- **Marking:** 2 marks for identifying the trade-off between vision and other senses/energy allocation.

Question 4 (Total 6 marks)

a. Calculate the allelic and genotypic frequencies. (2 + 2 marks)

- **Given:** $n = 1000$, Non-tasters (tt) = 360. Tasters (T_{-}) = 640.

- **Step 1: Calculate q (frequency of t allele)**

$$q^2 = \text{frequency of } tt = \frac{360}{1000} = 0.36$$

$$q = \sqrt{0.36} = 0.6$$

- **Step 2: Calculate p (frequency of T allele)**

$$p = 1 - q = 1 - 0.6 = 0.4$$

- **Step 3: Calculate genotypic frequencies under HWE**

$$p^2(TT) = (0.4)^2 = 0.16 \quad (\text{or } 160 \text{ individuals})$$

$$2pq(Tt) = 2 \times 0.4 \times 0.6 = 0.48 \quad (\text{or } 480 \text{ individuals})$$

$$q^2(tt) = 0.36 \quad (\text{or } 360 \text{ individuals})$$

- **Marking:**
 - **Allelic frequencies (2 marks):** 1 mark for $p = 0.4$, 1 mark for $q = 0.6$.
 - **Genotypic frequencies (2 marks):** 1 mark for $TT = 0.16$, 1 mark for $Tt = 0.48$ (and $tt = 0.36$ is implied but should be correct).

b. Which is the most common Phenotype? (1 mark)

- **Answer:** The most common phenotype is **taster** (640 out of 1000 individuals).
- **Marking:** 1 mark for “taster”.

c. Does the most common allele correspond to the most common phenotype? How do you explain this observation? (1 mark)

- **Answer:** No. The most common allele is t (frequency 0.6), but the most common phenotype is taster. This is because the recessive allele (t) is more common, but the dominant phenotype (taster) includes both TT and Tt individuals. The dominant allele (T) is less common but expresses in heterozygotes, making the taster phenotype more frequent.
- **Marking:** 0.5 marks for “No”, 0.5 marks for the explanation.

Question 5 (Total 5 marks)

a. State the “four postulates” of evolution with reference to the figure. (2 marks)

- **Answer:** The four postulates are:
 1. **Phenotypic Variation:** The figure shows variation in the female population (two tusks, one tusk, no tusks) in panel A.
 2. **Heritability of the Phenotype:** The trait is heritable – not clearly shown in the panel but implied in panel C where the offspring generation shows a similar pattern to the survivors, not the pre-war population.
 3. **Differences in survival/reproduction:** Shown in panel B where the three types of females have differential survival.
 4. **Survival/Reproduction non-random with respect to a phenotype:** Panel B shows that females with no tusks had the highest survival rate during the war, while those with two tusks had the lowest.
- **Marking:** Award 0.5 marks for each postulate correctly stated and linked to the figure. The key is the link to the figure.

b. Is this an example of Adaptive Evolution by Natural Selection? Back up your answer with reference to the figure. (2 marks)

- **Answer:** Yes. The figure shows a change in the population’s trait distribution over time due to an environmental pressure (war/poaching). Before the war (A), the population had a mix of tusk phenotypes. After the selective event (B), the frequency of tuskless females increased. This trait was then passed to the next generation (C), showing an adaptive shift towards tusklessness, which conferred higher survival.

- **Marking:** 1 mark for “Yes”. 1 mark for using the figure to explain the change in frequency and heritability.

c. What is the agent of selection in this case? (1 mark)

- **Answer:** The agent of selection was **human hunting** / **poaching** (specifically during the Mozambique civil war for ivory trade).
- **Marking:** 1 mark for “human hunting” or “poaching”.

Question 6 (Total 3 marks)

Calculate the number of gametes that are recombinants between A-B loci ONLY. (3 marks)

- **Given:** A-B distance = 10 cM, B-C distance = 20 cM. Order: A-B-C. Total gametes = 1000.
- **Step 1: Understand the question.** “Recombinants between A-B loci ONLY” means gametes that have a crossover between A and B, but *no* crossover between B and C.
- **Step 2: Calculate the frequency.**
 - The probability of a single crossover between A and B is $10\% = 0.1$.
 - The probability of *no* crossover between B and C is $1 - 0.2 = 0.8$.
 - The frequency of gametes that are recombinant for A-B only is the product of these two independent probabilities (assuming no interference): $0.1 \times 0.8 = 0.08$.
- **Step 3: Calculate the number.**

$$\text{Number} = 0.08 \times 1000 = \mathbf{80}$$

- **Marking:** Full marks for 80. Partial marks for correct approach but calculation error (e.g., using 0.2 for no crossover).